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Planetary Modes, Distance Uncertainty and the Limits of Visibility

A finite model-based scan and a quantitative observation-design framework

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Abstract

Can a fixed UNIQUANT anchor, planetary oscillations and observer corrections reveal a uniquely determined spatial geometry? This report performs a bounded first calculation rather than assuming the answer. A frozen dictionary of eight nominal planetary frequencies generates 72 first- and second-order combinations. Each is fitted separately to eight heliocentric JPL DE441 trajectories, giving 576 three-dimensional fits with chronological model-reference validation and test windows. The combined dictionary is severely ill-conditioned on the five-year training interval: at a specified numerical cutoff its 146-column design has rank 66. Selected outer-planet modes can have physically misleading frequency labels, demonstrating the limits of short-arc interpretation. A time-origin shift leaves paired-mode amplitudes invariant. Separate design calculations quantify diffraction, Gaussian-beam spreading and stereo depth uncertainty, including a conditional optimal-baseline result. The available inputs do not contain independent raw observations or observational covariance, so no physical significance levels, new distance scale or universal best observing frequency are inferred.

What was calculated. The deliverable includes every one of the 72 candidate definitions, all 576 single-candidate amplitude fits, selected models, source checksums, optical design tables and executable reproduction scripts.

What "significant" would require. A large fitted amplitude or a smaller reference-model approximation error is not an observational significance test. Physical significance requires independently measured quantities, their joint error model and a prespecified search/validation procedure.

Main answer. A finite observer-aware reconstruction programme is possible. Neither a time anchor nor a collection of sine waves alone determines a unique absolute "true geometry." A useful observing volume is defined by accuracy and detectability targets, not by a perfect frequency or an infinitely long hard ray cylinder.

1 Inputs, frames and the role of the jj anchor

The current UNIQUANT manuscript [1] defines

$$q = \frac{75}{1012}\,\mathrm{s}, \qquad t_n - t_{jj} = q(A_n + \rho_n), \qquad -\tfrac{1}{2} \le \rho_n < \tfrac{1}{2}. \tag{1}$$

The civil jj label is 13 July 2029, 00:00 CEST, or 12 July 2029, 22:00 UTC. This future reference is a definition, not an observation of a newly verified physical event. A precise UTC-to-TDB conversion needs an explicit time-scale convention, including future UTC adjustments. The scan avoids asserting that conversion: it uses an arbitrary TDB phase origin and tests origin invariance directly.

1.1 A time anchor does not supply a spatial reference frame

For a sinusoidal pair, changing t_0 to $t_0 + \Delta$ rotates the coefficient pair:

$$\mathbf{a}' = \mathbf{a}\cos\delta + \mathbf{b}\sin\delta, \qquad \mathbf{b}' = -\mathbf{a}\sin\delta + \mathbf{b}\cos\delta, \quad \delta = 2\pi f\Delta. \tag{2}$$

Consequently $\|\mathbf{a}\|^2 + \|\mathbf{b}\|^2$ and the singular values of $[\mathbf{a}\ \mathbf{b}]$ do not change. Fixing jj aligns phase bookkeeping; it does not create new amplitude information or statistical significance. Origin, axis orientation, length scale, observer pose and clock calibration remain separate choices or measurements.

1.2 Available trajectory data

The existing atlas data contain 4019 daily model states per object from 2020-01-01 to 2031-01-01 TDB, obtained from NASA/JPL Horizons DE441. The normalized input and original-query hashes are included. All nine raw source checksums were verified. These are fitted, numerically integrated ephemerides, not independent raw telescope observations [2, 3].

Positions are in AU and velocities in AU/day, with origin at the solar-system barycenter and fixed Ecliptic J2000 axes. The states are *geometric*, with `VEC_CORR=NONE`. Sun, Mercury, Venus and Earth are body centers; Mars through Neptune are planetary-system barycenters. This distinction is preserved in the files. Subtracting the same-epoch solar state produces the heliocentric vectors used below.

For geometric common-time vectors,

$$(\mathbf{r}_p - \mathbf{r}_E) + (\mathbf{r}_E - \mathbf{r}_S) = \mathbf{r}_p - \mathbf{r}_S. \tag{3}$$

The programme checks this identity. It is a frame conversion inside the supplied model. It does not recover an unknown observed universe, and it is not a light-time, aberration, refraction or gravitational-bending correction.

2 The finite family and the calculation

Let f_j be the eight nominal frequencies obtained from the mean longitude rates in Table 1 of the JPL approximation [4]. The Earth frequency in that table is for the Earth-Moon barycenter; the fitted Earth trajectory is body center 399. Constant mean-angle rates are a declared basis approximation, not exact true anomalies or a complete perturbation theory.

Use phases $\phi_j(t) = 2\pi f_j(t - t_0)$ and integer combinations

$$f_{\mathbf{k}} = |\mathbf{k} \cdot \mathbf{f}|, \qquad 0 < \|\mathbf{k}\|_1 \leq 2, \qquad \mathbf{k} \sim -\mathbf{k}. \tag{4}$$

There are 8 fundamentals, 8 second harmonics, 28 sums and 28 differences: 72 nominal candidates. Allowing order three would give 416, but those additional candidates were not tested. Without a finite order or bandwidth the request for "all combinations" has no finite endpoint.

2.1 Separate fits and geometric amplitudes

Each candidate is fitted separately to each planet's three-coordinate vector:

$$\widehat{\mathbf{r}}(t) = \mathbf{c} + \mathbf{d}\frac{t - t_c}{T_{\text{train}}} + \mathbf{a}\cos[2\pi f_{\mathbf{k}}(t - t_0)] + \mathbf{b}\sin[2\pi f_{\mathbf{k}}(t - t_0)]. \tag{5}$$

Linear least squares estimates the coefficient vectors. The scalar design has four columns and the three coordinates share that design. The reported isolated-mode amplitude is

$$A_{\text{rms}} = \sqrt{\frac{\|\mathbf{a}\|^2 + \|\mathbf{b}\|^2}{2}}. \tag{6}$$

It is the RMS vector amplitude over a complete cycle of that isolated sinusoidal pair, not necessarily its RMS over the short data window. The two singular values of $[\mathbf{a}\ \mathbf{b}]$ give its ellipse semiaxes. Separate fits are not unique amplitudes of simultaneously present close modes.

Window	TDB calendar labels	Samples
Fit	2020-01-01 to before 2025-01-01	1827
Selection	2025-01-01 to before 2028-01-01	1095
Final reference test	2028-01-01 to before 2031-01-01	1096

The candidate with minimum selection-window RMSE is retained per body, without using final-test error for selection. A constant-plus-linear fit is a deliberately simple comparison baseline, not a state-of-the-art orbit model. An own-frequency fundamental-plus-second-harmonic comparison is also reported in the data files. Future table entries are ephemeris predictions, not future observations. All these windows test model compression/extrapolation within the same reference source. Splitting table epochs does not independently split the observations used to construct the underlying DE441 solution.

3 Results: descriptive modes, not newly discovered forces

Body	Mode	P (d)	Train cycles	A_{rms} (AU)	Val./linear	Test/linear
Mercury	F:Mer	87.97	20.757	0.37921	0.10204	0.10216
Venus	F:Ven	224.70	8.126	0.72331	0.00335	0.00326
Earth	F:Ear	365.26	4.999	0.99981	0.00813	0.00770
Mars	F:Mar	686.98	2.658	1.52440	0.04255	0.04503
Jupiter	Jup+Nep	4041.86	0.452	4.90639	0.01916	0.04365
Saturn	F:Sat	10755.88	0.170	8.71141	0.02005	0.04938
Uranus	2Nep	30094.83	0.061	17.66063	0.00625	0.01302
Neptune	Ura+Nep	20324.87	0.090	2.82151	0.00938	0.03434

Table 1: Best separate candidate selected on the validation window. F denotes a body's nominal fundamental; Jup+Nep and Ura+Nep denote sums; 2Nep is Neptune's second harmonic. Ratios below one improve on a weak linear reference baseline. Amplitudes are conditional fit coefficients, not jointly identified physical modes.

The inner planets select their own nominal fundamentals. Outer-body selections include sums of Jupiter and Neptune, twice Neptune, and a sum of Uranus and Neptune. These labels must not be read as evidence that those bodies are driven by the named combination. The five-year training interval spans only a small fraction of their long cycles; trend and sinusoidal curvature can imitate one another over short arcs.

The full simultaneous design contains an intercept, a trend and both columns of all 72 modes: 146 columns. After dividing each column by its training RMS, the computed singular spectrum gives effective numerical rank 66 at relative cutoff 10^{-12}, or 61 at 10^{-8}. The smallest/largest singular-value ratio is about 1.88×10^{-18}. These are numerical conditioning diagnostics for the stated scaling and window, not a theorem of exact algebraic rank deficiency.

Twenty candidates span fewer than two training cycles; 17 span fewer than one cycle in the entire eleven-year interval. Sixty-seven have a dictionary neighbor within $1/T_{\mathrm{train}}$. Cycle counts and Rayleigh spacing are warning heuristics, not absolute impossibility results for a known-frequency, high-SNR estimation problem.

3.1 Checks specific to the proposed representation

Refitting the selected modes after a phase-origin shift of 123.456 days preserves their amplitudes and predictions to numerical precision, as (2) requires. This is why fixing jj cannot by itself separate hidden modes.

The coordinate carrier $1/q$ has exactly

$$\frac{86400}{75/1012} = 1\,165\,824 \tag{7}$$

cycles per nominal daily step. On an exact daily grid in that same model-time coordinate it aliases to a constant phase. This does not assert equal proper-time increments of a physical observer, nor detect a physical 13.4933 Hz cosmic oscillation. The 35 unequal phase classes are also not 35 independently measured physical frequencies.

No observational covariance or independent raw measurements were supplied. Thus this calculation provides no p-values, confidence intervals for physical orbital amplitudes, or evidence that the JPL distance scale is incorrect.

4 Mode diagnostics

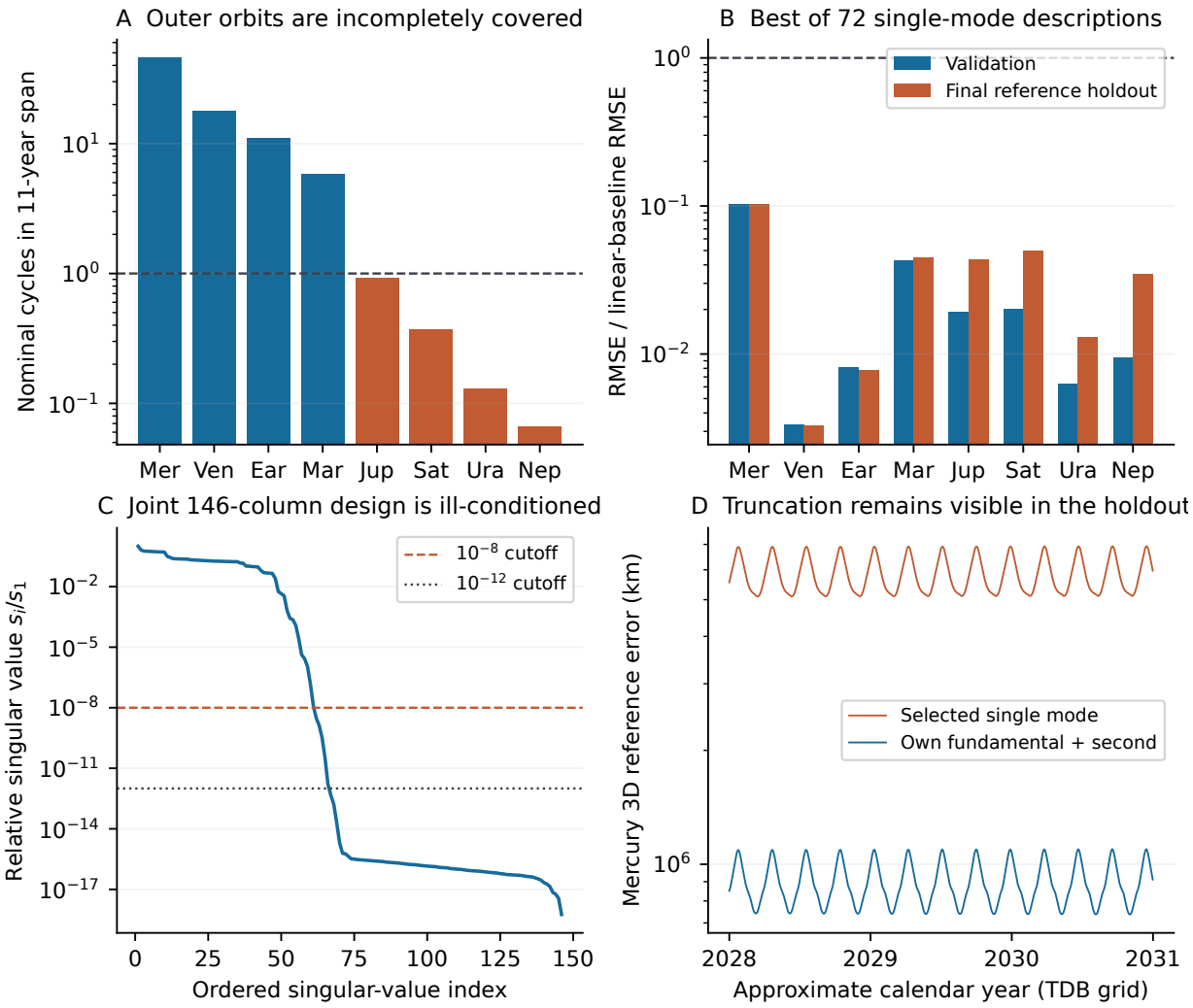


Figure 1: Reproducible diagnostics of the reference-model screen. Cycle coverage, selection/test error ratios, the joint singular spectrum and an example of harmonic truncation illustrate both representability and conditioning limits. The errors are relative to the supplied DE441 trajectory, not residuals of raw astronomical observations. The source bundle supplies the full tables and labels.

An improvement over a linear baseline need not be a precise astronomical result. The selected Mercury one-mode test RMSE is 0.03916 AU, about 5.86 million km. Its own fundamental-plus-second-harmonic approximation gives 0.006008 AU, about 0.899 million km. These are truncation/model-compression errors, not DE441 uncertainties or measured telescope errors. Absolute errors and ratios are both recorded in the source files.

The frequency dictionary was fixed from a published reference table. The scan therefore asks how that known frequency family describes the supplied model. A new physical frequency search would additionally require a specified continuous search range, treatment of aliasing, correlated noise and the selection penalty.

5 Distance uncertainty belongs inside the observation model

"Visibility error" contains several different problems: uncertainty of direction, uncertainty of distance, calibration bias, and whether photons are detectable. They cannot be represented by a single universal correction factor.

For a raw observation, use a forward model of the form

$$y_{ib} = H_b[X(t_{\mathrm{em}};\beta),\, z_b(t_{\mathrm{det}}),\, R_b,\, \mathrm{clock},\, \nu_i;\kappa] + \epsilon_{ib}, \qquad \mathrm{Cov}(\epsilon) = C. \tag{8}$$

The parameters include dynamics, mode coefficients, distances, observer pose and instrument calibration. In flat vacuum a signal additionally satisfies $c(t_{\mathrm{det}} - t_{\mathrm{em}}) = |\mathbf{x}(t_{\mathrm{em}}) - \mathbf{b}(t_{\mathrm{det}})|$. Real optical reconstruction may require aberration, gravitational propagation and atmospheric terms. Horizons explicitly distinguishes geometric and apparent quantities [2]; applying the same correction twice would be an error.

5.1 Local stereo sensitivity

For an ideal symmetric stereo geometry, with baseline B and midpoint-to-target distance r perpendicular to the baseline,

$$\varpi = 2\arctan\frac{B}{2r}, \qquad r = \frac{B}{2}\cot\frac{\varpi}{2}. \tag{9}$$

The local Jacobian for inputs (B, ϖ) is

$$J = \left(\frac{r}{B},\, -\frac{r^2}{B} - \frac{B}{4}\right), \qquad \sigma_r^2 \simeq J\Sigma J^T. \tag{10}$$

For $B \ll r$, negligible baseline error and angular noise small compared with the parallax, this reduces to $\sigma_r \simeq r^2\sigma_\varpi / B$. Small or noisy parallax requires inference in the original measurement space rather than blindly inverting an uncertain number. Common pose, baseline and clock errors are correlated across targets and must remain in the joint model.

After removing coordinate freedoms, inspect the whitened sensitivity matrix

$$S = C^{-1/2}\frac{\partial H}{\partial \xi}. \tag{11}$$

Small singular values identify locally weakly constrained parameter combinations. A nonsingular, well-conditioned local problem does not by itself establish global uniqueness. Priors can stabilize a fit but add assumptions; they are not new data. The current inputs lack the raw observation/covariance package needed to perform this joint physical reconstruction numerically.

6 A usable observing volume instead of a perfect frequency

Specify target detail ℓ_*, depth uncertainty σ_* and minimum SNR S_*. A useful observation region can then be defined by

$$\mathcal{V}_{\text{use}} = \{\mathbf{x} : \text{SNR} \geq S_*,\ \ell_\perp \leq \ell_*,\ \sigma_r \leq \sigma_*,\ \text{adequate viewing geometry and coverage}\}. \tag{12}$$

Its size depends on aperture, wavelength, source spectrum, transmission, exposure, detector, background, observer calibration and the requested information.

For an ideal circular aperture, the Rayleigh angular resolution is

$$\theta_R \simeq 1.22\frac{\lambda}{D}, \qquad \ell_\perp(r) \simeq r\theta_R. \tag{13}$$

This is a two-source resolution criterion [5]. It is not the centroiding precision used as σ_ϖ in a particular astrometric measurement. With $D = 5\,\text{cm}$ and $\lambda = 550\,\text{nm}$, the ideal value is 2.768 arcseconds: about 13.42 km detail at one million kilometres, or 2008 km at one AU. These are illustrative ideal limits, not a claim that all such targets are bright enough to detect.

6.1 Finite free beams spread

A separate ideal Gaussian-beam experiment gives [6]

$$w(z) = w_0\sqrt{1 + (z/z_R)^2}, \qquad z_R = \frac{\pi w_0^2}{\lambda}. \tag{14}$$

Here w is the $1/e^2$ intensity radius, not a hard cylinder edge. A waist radius $w_0 = 2.5\,\text{cm}$ at 550 nm has $z_R = 3.570\,\text{km}$ and $w(100\,\text{km}) = 0.7007\,\text{m}$. This waist example must not be identified with the hard 5-cm telescope aperture in (13).

6.2 Transmission and photon statistics

For direct-beam transmission, $T_\nu = \exp[-\int \mu_\nu\,\mathrm{d}s]$. With spectral flux F_ν, effective collecting area A_{eff} and exposure t, a suitable counting model has

$$N_s = t\int \frac{F_\nu A_{\text{eff}}(\nu)T_\nu}{h\nu}\,\mathrm{d}\nu, \qquad \text{SNR} \simeq \frac{N_s}{\sqrt{N_s + N_b + \sigma_{\text{read}}^2}}. \tag{15}$$

Noise terms and units must match the detector [7]. A shorter wavelength may improve diffraction while worsening transmission, background or optical efficiency. No material distribution, source luminosity or detector budget was specified here, so a numerical universal best frequency or detection horizon cannot be calculated from the supplied inputs.

7 How far should the next observer be placed?

For a target range r, desired relative depth standard uncertainty ϵ and assumed disparity precision σ_ω, the small-baseline design estimate is

$$B \gtrsim \frac{r\sigma_\omega}{\epsilon}. \tag{16}$$

For the hypothetical $\sigma_\omega = 1\,\mu$rad and $\epsilon = 1\%$, the required baselines are about 100 km at a range of one million km, 14,960 km at one AU, and 20.6 AU at one parsec. These figures assume independently calibrated geometry and a precision that does not degrade with range. They are not instrument specifications and are unrelated to simply substituting the Rayleigh criterion for an angular measurement uncertainty.

The exact angular sensitivity from (9) is

$$g(B) = \frac{r^2}{B} + \frac{B}{4}, \qquad g'(B) = -\frac{r^2}{B^2} + \frac{1}{4}. \tag{17}$$

At fixed r, with baseline-independent angular uncertainty and negligible baseline error, g is minimized at $B = 2r$. The sightlines then intersect at 90°. This conditional optimum is not a universal observer spacing: changing the baseline also changes source distances, brightness, overlap, occultation, clock and pose requirements in a real experiment.

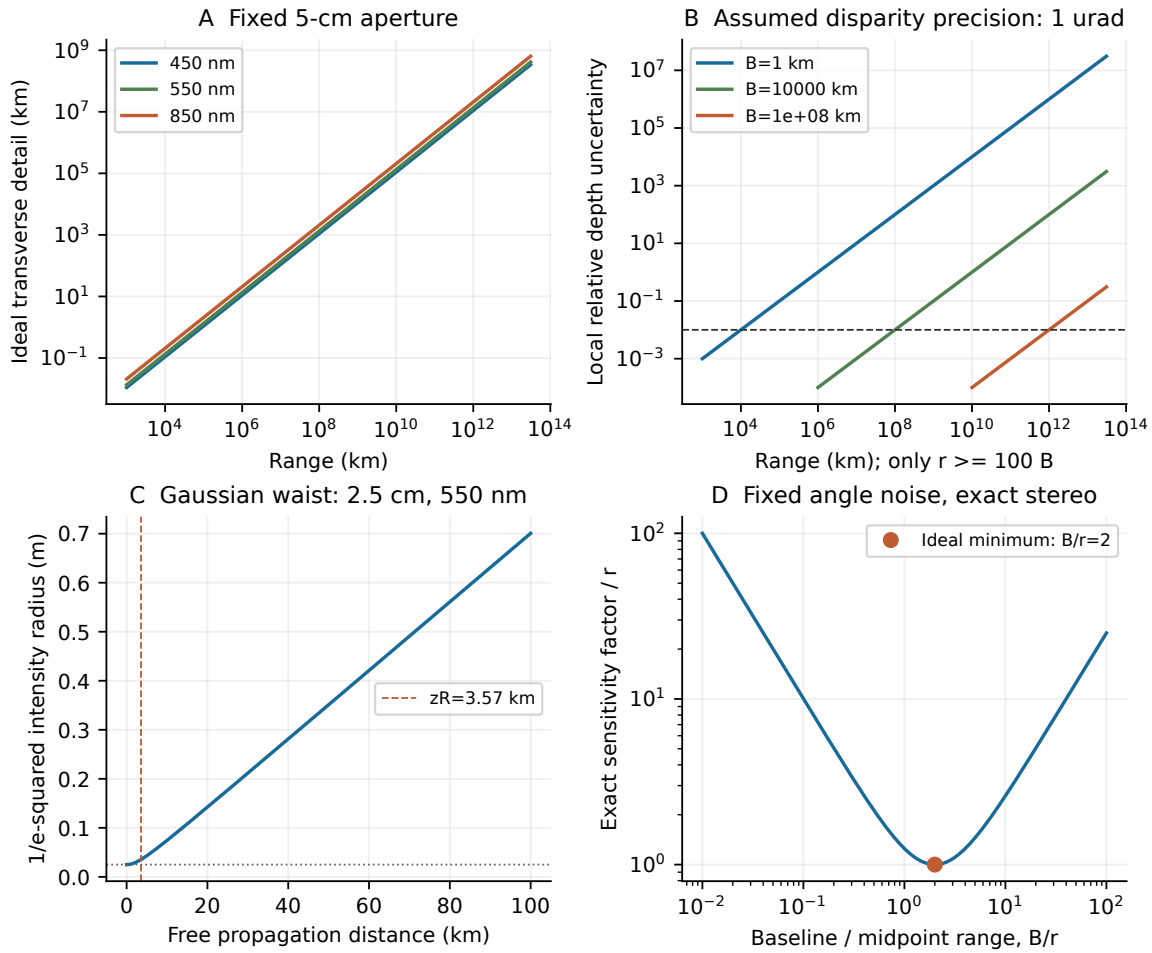


Figure 2: Ideal design sensitivities. The assumed parallax precision is independent of diffraction resolution. The Gaussian waist is a separate optical setup. No source/detector SNR budget is inferred. All plotted quantities are reproducible design calculations, not measured uncertainties.

8 What a physical reconstruction would need next

The existing calculation establishes a finite catalogue, a descriptive reference-model screen, explicit conditioning problems and ideal observation-design relations. It does not determine a revised three-dimensional map of everything previously observed. More frequency terms alone cannot supply unmeasured depth or remove an unknown common observer error.

At fixed finite aperture and nonzero wavelength, transverse detail grows with range. At fixed finite baseline and nonzero disparity uncertainty, depth uncertainty also grows without bound. Infinite-range perfect spatial resolution therefore does not follow from a finite design. Multiple viewpoints add information through their actual calibrated coverage and measurement operators.

1. Select one bounded target, time interval and required spatial/depth precision.
2. Obtain raw angles, calibrated range or timing data, observing-site/spacecraft states, frequency bands, clock/pose calibration and the joint error model.
3. Declare the spatial origin, axes, length scale and time convention. Map the jj definition consistently; do not interpret an arbitrary phase origin as new data.
4. Fit the physical state and observation/calibration parameters jointly. Check identifiability before interpreting individual amplitudes.
5. Compare a baseline observation/dynamics model with a prespecified added-mode model on withheld observations and viewing geometries. Correct for the full search, including any choices of frequencies, anchor or windows made after inspecting data.
6. Propagate covariance, posterior uncertainty and specified systematic-error scenarios. For correlated time series, null simulations must preserve the relevant sampling and error correlations and repeat the complete selection procedure.

The source bundle includes `candidate_modes.csv` (72 definitions), `amplitude_screen.csv` (576 fits), `body_summary.csv`, `reference_data.json`, `source_manifest.json`, the optical design tables, figures and scripts. Running `analyze.py` uses the included normalized data and verifies its checksum; it does not need a network call. Run `visibility.py` for the ideal design examples, then compile this report twice.

Interpretive boundary. A sinusoidal combination can describe a geometric projection, a nonlinear harmonic, an existing perturbation or an approximation of an incomplete arc. It is not automatically an energy flow or a newly discovered interaction. A fitted coordinate transformation is physically informative only where the data and stated assumptions constrain it.

Sources

References

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[7] Space Telescope Science Institute, *ACS Instrument Handbook*, Section 9.3, Computing Exposure Times. https://hst-docs.stsci.edu/acsihb/chapter-9-exposure-time-calculations/9-3-computing-exposure-times.